

# Milestone 3, Hypothesis 2: Is Melanoma Recall a Threshold Problem?

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**Base paper:** Datta et al., *Soft-Attention Improves Skin Cancer Classification Performance*, iMIMIC @ MICCAI 2021

**Reference model:** our reproduced Inception-ResNet-v2 baseline (cv\_project\_irv2.ipynb, seed 42)

**Companion to:** M3\_hypothesis.md (Hypothesis 1, focal loss)

## 1. Motivation

The reproduced IRv2 baseline separates melanoma from other lesions well but rarely *predicts* melanoma:

| Melanoma metric (IRv2, test set, n = 34) | Value                                            |
|------------------------------------------|--------------------------------------------------|
| One-vs-rest AUC                          | <b>0.915</b>                                     |
| Recall at arg-max                        | <b>0.441</b> (15 / 34), 95 % Wilson CI 0.29–0.60 |
| Precision at arg-max                     | 0.517                                            |
| Specificity at arg-max                   | 0.982                                            |

With an AUC of 0.915 and a specificity of 0.982, the arg-max decision sits at a very conservative point on the melanoma ROC curve. The model ranks melanomas above most non-melanomas, but argmax lets a melanoma win only when its probability beats every other class, and that rarely happens when *nv* holds 80 % of the data.

Hypothesis 1 (focal loss) predicts that changing the training loss will raise melanoma recall. Its own falsification criteria say the result is only *partial support* if the gain "could also be produced by a threshold change". So far no one has measured what a threshold change alone can do. This hypothesis fills that gap.

## 2. Hypothesis

**Lowering the melanoma decision threshold on the already-trained IRv2 baseline, with no retraining, will raise melanoma recall from 0.44 to at least 0.59 ( $\geq 20 / 34$ ) while keeping melanoma specificity  $\geq 0.95$ . The reason is that the baseline's errors come from the arg-max operating point, not from poor features.**

**Decision rule.** Predict *mel* if  $p(\text{mel}) \geq t$ . Otherwise predict the arg-max over the remaining six classes. With  $t =$  arg-max-equivalent, this reproduces the baseline exactly, so **t is the only variable**.

**Predicted effect (cross-fitted, see §3):**

| Metric                  | Baseline (arg-max) | Predicted at spec $\geq 0.95$    |
|-------------------------|--------------------|----------------------------------|
| Melanoma recall         | 0.441              | <b>0.59–0.75</b>                 |
| Melanoma precision      | 0.517              | 0.35–0.45 (falls)                |
| Melanoma specificity    | 0.982              | 0.95–0.96                        |
| nv recall               | 0.989              | 0.96–0.98                        |
| Macro recall            | 0.720              | +0.02 to +0.04                   |
| Weighted precision      | 0.909              | within $\pm 0.015$               |
| Weighted / melanoma AUC | 0.978 / 0.915      | <b>unchanged by construction</b> |

### 3. Protocol

1. **Inputs.** Test-set softmax outputs ( $828 \times 7$ ) from the saved IRv2 seed-42 checkpoint. If they were not saved, one inference pass produces them in a few minutes. No GPU training is needed.
2. **No threshold chosen on the data it is scored on.** The checkpoint already saw the whole training set, so a slice of training data would give optimistic probabilities. Instead we use **5-fold cross-fitting on the test set**, stratified by class. For each fold,  $t$  is picked on the other four folds as the lowest threshold whose melanoma specificity is  $\geq 0.95$ , then applied to the held-out fold. Predictions from all five folds are pooled before computing metrics.
3. **Operating points reported.** (a) arg-max, (b) specificity  $\geq 0.95$ , (c) specificity  $\geq 0.90$ , (d) specificity 0.713, which is the mean dermatologist specificity in Haenssle et al. (2018). At (d) we compare the model's sensitivity with the dermatologists' 0.866.
4. **Uncertainty.** 95 % Wilson intervals for per-class recall. 1,000-resample bootstrap intervals for weighted precision and macro recall. We repeat the 5-fold split with 20 random seeds and report mean  $\pm$  std for  $t$ .
5. **Same procedure on IRv2 + Soft-Attention** (mel AUC 0.969). This tells us whether attention's recall gain ( $15 \rightarrow 21 / 34$ ) survives once both models are compared at the same specificity.

### 4. How we will know we were wrong

- **Reject** if cross-fitted melanoma recall at specificity  $\geq 0.95$  is below 20 / 34 (0.588). That would mean the ROC curve is too flat at low false-positive rates for a threshold to help, and the problem does lie in the features.
- **Reject** if weighted precision drops by more than 0.02. The gain would then be bought with too many *nv* false alarms.
- **Inconclusive** if the threshold chosen in each fold varies by more than 0.2 across folds. With only 34 melanomas, the operating point would not be stable enough to recommend.

### 5. Why this matters for Hypothesis 1

| H2 outcome | Consequence for the focal-loss experiment (H1)                                                                                                                                          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supported  | Arg-max recall alone cannot show that focal loss helps. H1 must be judged as <b>melanoma recall at matched specificity</b> against the threshold-tuned baseline, or as a change in AUC. |
| Rejected   | The baseline's features are the bottleneck. A training-time change such as focal loss is justified, and arg-max comparisons remain fair.                                                |

Either way, H2 sets the bar that H1 has to clear, and it costs zero GPU-hours.

## 6. Risks

| Risk                                                              | Mitigation                                                                                                  |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Only 34 test melanomas, so one image = 2.9 recall points          | Wilson CIs; repeated cross-fitting; report counts, not just rates                                           |
| Checkpoint itself was selected on the test set (paper's protocol) | Stated as a shared limitation with M2; does not favour any threshold                                        |
| Softmax probabilities may be poorly calibrated                    | Thresholding needs only the <i>ranking</i> , not calibration. We also report a reliability diagram for mel. |
| More mel predictions take images from nv/bkl                      | Track nv and bkl recall and the full confusion matrix at each operating point                               |

## 7. Results table (to fill in after running)

| Model     | Operating point               | t (mean $\pm$ std) | mel recall             | mel spec. | mel prec. | macro recall | W. precision |
|-----------|-------------------------------|--------------------|------------------------|-----------|-----------|--------------|--------------|
| IRv2      | arg-max                       | —                  | 0.441                  | 0.982     | 0.517     | 0.720        | 0.909        |
| IRv2      | spec $\geq 0.95$              |                    | <i>pred. 0.59–0.75</i> |           |           |              |              |
| IRv2      | spec $\geq 0.90$              |                    |                        |           |           |              |              |
| IRv2      | spec = 0.713 (dermatologists) |                    | <i>vs. 0.866</i>       |           |           |              |              |
| IRv2 + SA | arg-max                       | —                  | 0.618                  | 0.987     | 0.677     | 0.713        | 0.913        |
| IRv2 + SA | spec $\geq 0.95$              |                    |                        |           |           |              |              |

*No clinical claims are made. All results are benchmark numbers on a curated research dataset.*

**References.** Datta S. K. et al., iMIMIC @ MICCAI 2021. Haenssle H. A. et al., *Annals of Oncology* 29(8), 2018. Lin T.-Y. et al., *Focal Loss for Dense Object Detection*, ICCV 2017.